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Dependence of the Lifelength of a Hydraulic Breaker Pick on the Wear Resistance of Its Material

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Abstract

The paper presents experimental results on the effect of blow energy on the percussion–abrasion wear intensity of a model tool in the form of a conical pick, made of three grades of tool steel, during percussive interaction with a granite block. Part of the tools was subjected to the standard heat treatment practised at pick manufacturing plants, while the other part received additional cold treatment or cryogenic treatment. To justify the transfer of the experimental data to full-scale conditions, the operating parameters of the experiment were set so that its conditions were similar to the working conditions of a hydraulic breaker pick. To achieve this similarity, the level of contact stresses between the model tool and the rock in the experiments was the same as in percussive penetration of the pick. A relationship between the percussion–abrasion wear intensity and the blow energy of the model tool was established, and on this basis the wear intensities of the pick of a JCB HM380 hydraulic breaker during the breaking of granite blocks were calculated. The pick lifelength was evaluated, taken as the number of blows delivered by the pick until complete wear of its tip. From a comparison of the lifelength of picks made of the steels studied and subjected to the above types of heat treatment, it is concluded that it is expedient to introduce cold treatment and cryogenic treatment operations into the manufacturing process of hydraulic breaker picks.

Keywords: *hydraulic breaker pick, percussive action, blow energy, granite, percussion–abrasion wear, contact stress, tip dulling, cold treatment, cryogenic treatment, pick lifelength.*

Introduction

A considerable number of works, for example [1–14], are devoted to the study of the interaction of a percussive tool with the rock being broken and to methods of increasing its wear resistance.

In the authors' work [15], experimental results were presented on the wear of the blunt (dulled) area of a cone-shaped model tool as a result of its repeated percussive penetration into a granite block. It was found that the obtained dependences of the tool mass loss on the number of blows are described by an inverse power function of the form $\Delta m = An^{-B}$, where A and B are coefficients depending on the material and type of heat treatment of the tool, with 3 stages, each differing in the amount of metal mass loss and the rate of its decrease with increasing number of blows. Using the presented mathematical model, it was shown that the transition from one wear stage to another is determined by the ratio between the stresses reached at the contact between the tool and the rock and the ultimate strength and yield strength of the tool material exhibited under dynamic conditions. At the final stage of wear, which is the longest, the metal mass loss per blow does not change

significantly with the number of blows and can serve as an indicator of the percussion–abrasion wear intensity of a tool made of the given steel, K_M [mg per blow].

In the present paper, the dependence of K_M [15] on the blow energy A_M of the model tool is established and used to estimate the percussion–abrasion wear intensity $K_П$ of a hydraulic breaker pick and its lifelength (number of blows) until complete wear of the tip.

[Figure 1 — see original PDF, page 6] General view and schematic diagram of the experimental setup for studying the wear patterns of a model percussive tool: 1 — rock block; 2 — model percussive tool; 3 — striker; 4 — roller; 5 — cam; 6 — reduction gear; 7 — electric motor

Experimental Procedure

The general view and the diagram of the experimental setup are shown in Fig. 1.

The model tool 2 (see Fig. 1), in the form of a conical pick with a tip angle β = 30° and a cylindrical-part radius r_M = 4 mm (required to satisfy the similarity criterion), was weighed on an analytical balance (±0.1 mg) and then fixed in the massive striker 3 of the setup.

[Figure 2 — see original PDF, page 6] Surface of the granite block after a series of blows by the model tool

A cylindrical granite core sample 1 from the Vyborg massif, with a hardness of f = 13 on the M.M. Protodyakonov scale and a uniaxial compressive strength $\sigma_{сж}$ ~200 MPa, was placed under the tool. Tool 2 together with striker 3 fell freely from a given height (h_M = 42 mm), striking the surface of the sample with its tip. The pre-impact velocity of the tool, calculated from h, was 0.91 m/s. With the same model tool, a series of n blows was delivered to the surface of sample 1, each subsequent blow being displaced by a distance ≥ $3d_Л$, where $d_Л$ is the diameter of the crater (Fig. 2).

After each blow, the mass of the model tool was measured to determine its loss Δm due to the blow. With a large number of blows (over 40), Δm per blow was determined over an interval of blows (n = 10–100) as the quotient of the total value of the function over n blows divided by the number n in the interval. By varying the mass of striker 3 (2.8; 4.2; 5.6 and 7.0 kg), the effect of the kinetic energy of the blow (A_M = 1.16; 1.74; 2.32 and 2.90 J) on the form of the dependence of tool mass loss Δm on the number of blows n was investigated.

Steels 38ХМ, У8А and Х12МФ were chosen as model tool materials. The first is a typical material of picks of domestic (Russian) and foreign (AISI P20 steel) manufacture. The second is the material of percussive teeth of active-action excavator buckets, which performed satisfactorily in tests [16], and the third is a material with high hardness and toughness in the hardened state, which has undergone field tests [16] as a hardfacing to prevent tooth wear.

Part of the tools made of these steels was subjected to the standard heat treatment (TTO) practised at pick manufacturing plants — oil quenching at 800–1000 °C followed by tempering at 200 °C; the other part received additional cold treatment (holding after quenching at minus 75 °C for 5 h) or cryogenic treatment (holding after quenching for 20 h in liquid nitrogen at minus 196 °C), as operations widely used to increase the wear resistance of metal-cutting tools [17–26].

Justification of the Similarity of the Penetration Processes of the Model Tool and the Hydraulic Breaker Pick into Rock

To justify the transfer of the experimental data obtained in the laboratory with the model tool to full-scale conditions, the operating parameters of the experiment were set so that its conditions were similar to the working conditions of the hydraulic breaker pick.

Based on the conclusion of K.I. Ivanov [27], it was assumed that, to ensure similarity of the conditions of penetration of the model percussive tool and the pick into rock, the experiment must provide a level of contact stresses between the model tool and the rock close to that occurring during percussive penetration of the pick. This guarantees the same cyclic durability of the tool and unchanged energy consumption for breaking a unit volume of rock.

The physical equation of the process of penetration of the tool and the pick into rock was represented as the functional

$$\varphi = (A, E_{ME}, \mu_{ME}, r, \beta, \sigma_k, E_{\Pi}, \mu_{\Pi}), \quad (1)$$

where A, E_{ME}, μ_{ME}, r, β, σ_k, E_{Π}, μ_{Π} are the blow energy, the modulus of elasticity and Poisson's ratio of the tool, the radius of the cylindrical part and the tip angle of the model tool and the pick; the contact stress; and the modulus of elasticity and Poisson's ratio of the rock.

Dimensions of the parameters determining the penetration process:

$$[\beta] = [L^0M^0T^0] \Rightarrow P_5,$$
$$[\sigma_k] = [L^{-1}M^1T^{-2}] \Rightarrow P_6,$$
$$[E_{\Pi}] = [L^{-1}M^1T^{-2}] \Rightarrow P_7,$$
$$[\mu_{\Pi}] = [L^0M^0T^0] \Rightarrow P_8,$$
$$[A] = [L^2M^1T^{-2}] \Rightarrow P_1,$$
$$[E_{ME}] = [L^{-1}M^1T^{-2}] \Rightarrow P_2,$$
$$[\mu_{ME}] = [L^0M^0T^0] \Rightarrow P_3,$$
$$[r] = [L^1M^0T^0] \Rightarrow P_4, \quad (2)$$

where L is length; M is mass; T is time.

Thus, the complete dimensional matrix had a size of n×q, where n = 8 is the total number of parameters determining the penetration process and q = 3 is the number of basic units of measurement of the parameters.

After excluding from (2) the dimensionless parameters (P_3, P_5 and P_8) and reducing parameters with identical dimensions to one (in our case, of the parameters P_2, P_6 and P_7 it is expedient to keep P_6), a reduced matrix of the following form was obtained

$$\|A\| = \begin{matrix} \| 2 & 1 & -2 \| \Rightarrow A \\ \| -1 & 1 & -2 \| \Rightarrow \sigma_k \\ \| 1 & 0 & 0 \| \Rightarrow r \end{matrix} \quad (3)$$

The total number of similarity criteria N_{π}, determined as N_{π} = m – l (where m and l are the total number of parameters (A, σ_k and r) and the number of independent parameters (A and r)), is 3 – 2 = 1 in our case. The specific expression of the similarity criterion π for the independent parameters A and r had the form

$$\pi = P_3(P_1^x, P_2^y)^{-1} = \sigma_k(A^x, r^y)^{-1}, \quad (4)$$

where x and y are unknown exponents of the independent parameters to be determined to reveal the specific form of the similarity criterion.

Since the similarity criterion is a dimensionless quantity, the dimensions of the numerator and the denominator must be the same, and from relation (4) we wrote

$$[\sigma_k] = [A]^x[r]^y. \quad (5)$$

Expanding the dimensions of the parameters of the last expression

$$[L^{-1}M^1T^{-2}] = [L^2M^1T^{-2}]^x[L^1M^0T^0]^y \quad (6)$$

and solving the resulting system of equations

$$\begin{gathered} 2x + y = -1 \\ x = 1 \\ -2x = -2 \end{gathered} \quad (7)$$

we obtained x = 1; y = −3.

Thus, the similarity criterion for the modelling conditions ($\pi_М$) and the pick working conditions ($\pi_П$) had the form

$$\pi = idem \Leftrightarrow \pi_M = \pi_П \Leftrightarrow \sigma_{k,M}A_M^{-1}r_M^3 = \sigma_{k,П}A_П^{-1}r_П^3 \quad (8)$$

Hence, to ensure equality of $\sigma_К$, we obtained

$$\sigma_{k,M} / \sigma_{k,H} = (A_M / A_П)\cdot(r_M / r_П)^{-3}, \quad (9)$$

where $A_М$ and $r_М$; $A_П$ and $r_П$ are the blow energies and cylindrical-part radii of the model tool and of the hydraulic breaker pick, respectively.

Having chosen the drop height of the model tool $h_М$ = 42 mm as convenient for the experiment, we obtained a blow energy $A_М$ = 1.16 J (for a striker mass of 2.8 kg). After substituting $A_М$ into (9) together with the values $A_П$, $r_П$ of the hydraulic breaker pick ($A_П$ = 981 J, $r_П$ = 39 mm for the JCB HM380 hydraulic breaker), the cylindrical-part radius of the model tool $r_М$ = 4 mm, at which the similarity condition is satisfied, was established.

Thus, to model the conditions of pick penetration into rock, the drop height $h_М$ of the tool and its radius $r_М$ should be 42 mm and 4 mm, respectively, which was implemented in the experiments.

As for ensuring equality of the energy consumption for rock breaking, a comparison of this indicator for the experimental conditions (which, based on the data of [29], can be estimated at 14 MJ/m^3) with the corresponding indicator for the hydraulic breaker (10.4 MJ/m^3 [16]) showed that their values are close.

Consequently, when the established values of $h_М$ and $r_М$ are ensured in the experiment, force and energy similarity of the systems "model percussive tool" – "rock" and "pick" – "rock" is maintained.

[Figure 3 — see original PDF, page 9] Dependence of the total mass loss of the model tool made of 38XM steel on the number of blows n at different blow energies: 1 — $A_{М1}$ = 1.16 J; 2 — $A_{М2}$ = 1.74 J; 3 — $A_{М3}$ = 2.32 J; 4 — $A_{М4}$ = 2.90 J

Experimental Results

The results of experiments on the effect of blow energy A_M on the mass loss of the model tool and its percussion–abrasion wear intensity K_M are illustrated in Fig. 3 using the example of a tool made of 38XM steel.

It turned out that, as the blow energy A_M increases, the curves $\Sigma\Delta m = f(n)$ for the model tool made of all steels shift upward (Fig. 3), and the percussion–abrasion wear intensity K_M increases. The values of K_M and A_M are in direct proportion: when A_M increases n times, K_M increases practically by the same factor. For example, when the blow work increases from A_{M1} = 1.16 J to A_{M4} = 2.9 J, i.e. by a factor of 2.5, the ratio of wear intensities K_{M4}/K_{M1} is 2.41 for 38XM steel, 2.63 for У8А and 2.37 for Х12МФ (Fig. 4), i.e. the following equality is practically observed

$$K_{Mi} = K_{M1} \cdot A_{Mi} / A_{M1} \quad (10)$$

Since the similarity criterion (9) was satisfied in the experiments, it was assumed that the established relationship (10) between K_M and A_M can be used to determine the percussion–abrasion wear intensity of the hydraulic breaker pick $K_П$ from the wear intensity K_M of a model tool made of the corresponding material and the ratio of the blow energies of the pick $A_П$ and the tool A_M

$$K_П = K_M \cdot A_П / A_M \quad (11)$$

[Figure 4 — see original PDF, page 9] Relationship between K_{Mi}/K_{M1} and A_{Mi}/A_{M1} for tools made of steels 38XM (●), У8А (■) and Х12МФ (▲)

Determination of the Percussion–Abrasion Wear Intensity of the Hydraulic Breaker Pick and Its Lifelength

[Figure 5 — see original PDF, page 10] Adopted scheme of the change in the dulling radius of the pick as a result of wear

It was assumed that the limit state of the pick (end of its lifelength) occurs upon complete wear of its tip, i.e. when the radius $r_{сф}$ of the hemispherical dulling of the pick tip becomes equal to the radius of its cylindrical part

$$r_{сфк} = r_n \quad (12)$$

Assuming that the overwhelming part of the worn metal comes from the contact area of the pick with the rock, the number of blows N^* that the pick must deliver to the granite block for the radius $r_{сф}$ to change from $r_{сф0}$ = 13 mm (dulling radius of the initial pick) to $r_{сфк} = r_П$ = 39 mm was calculated, owing to the reduction of the hemisphere volume at each blow by the amount v' corresponding to $K_П$,

$$v' = K_П / \rho \quad (13)$$

where ρ is the density of the pick material ($7.85 \cdot 10^3$ kg/m^3 for all steels) (Fig. 5). The established value N^* was taken as the pick lifelength.

Table. Percussion–abrasion wear parameters of the model tool and of the JCB HM380 hydraulic breaker pick during penetration into granite, compared with the pick lifelength

Steel grade and type of pick heat treatment	K_M, mg	$K_П$, mg	v', mm^3	$N^* \cdot 10^{-5}$
38XM$_{ТТО}$	0.0027	2.28	0.29	4.3

Steel grade and type of pick heat treatment	$K_м$, mg	$K_п$, mg	v', mm^3	$N^*\cdot10^{-5}$
38ХМ$_х$	0.0023	1.94	0.25	5.1
38ХМ$_{кр}$	0.0021	1.74	0.22	5.7
У8А	0.0019	1.61	0.20	6.1
У8А$_х$	0.0017	1.44	0.18	6.8
У8А$_{кр}$	0.0016	1.31	0.17	7.5
Х12МФ	0.0013	1.14	0.15	8.6
Х12МФ$_х$	0.0010	0.85	0.11	11.6
Х12МФ$_{кр}$	0.0007	0.57	0.07	17.3
ТТО — standard heat treatment; х — cold treatment; кр — cryogenic treatment of the pick.				

The values of $K_м$ of the model tool established in the experiments (for $A_{м1}$ = 1.16 J), the wear intensities $K_п$ calculated from them by (18) (for $A_п$ = 981 J), the volumes v' corresponding to $K_п$, and the established values of N^* are presented in the table.

As can be seen from a comparison of the data in the table, if the traditionally used pick material, 38ХМ steel, is replaced with more wear-resistant materials — У8А and Х12МФ steels — the tool lifelength will increase from $4.3\cdot10^5$ to $6.1\cdot10^5$ and $8.6\cdot10^5$ blows, respectively, i.e. by factors of 1.4 and 2.

Cold treatment and cryogenic treatment of tools made of all the steels considered can increase by a factor of 1.1 to 2 the number of blows that the pick can deliver before complete dulling, compared with a similar pick after standard heat treatment.

This positive effect of cold treatment is explained by the fact that holding tools made of the high-carbon steels У8А (~0.8% C) and Х12МФ (~1.5% C) at sub-zero temperatures leads to a significant reduction in the content of "soft" retained austenite in their structure, where it remained in considerable amounts after quenching. It also promotes an increase in the carbide content in the structure of tools made of the alloy steels 38ХМ and Х12МФ. Moreover, the lower the cold treatment temperature, the higher its effectiveness [29].

Main Conclusions

- A laboratory test procedure is presented for studying the wear patterns of a model percussive tool made of various steels with various heat treatments under conditions simulating the wear conditions of a hydraulic breaker pick during its penetration into rock.
- Tests of the model tool made of three grades of tool steel showed that its percussion–abrasion wear intensity $K_м$ during penetration into a granite block increases with blow energy in direct proportion.
- Based on the established wear intensities $K_м$ of the model tool and the form of their dependence on blow energy, the percussion–abrasion wear intensity $K_п$ and the lifelength of a JCB HM380 hydraulic breaker pick made of three steel grades and subjected to three types of heat treatment were calculated for penetration into granite.
- It is shown that if the traditionally used pick material, 38ХМ steel, is replaced with more wear-resistant materials — У8А and Х12МФ steels — the number of blows the pick can deliver before complete wear of its tip (pick lifelength) will increase by factors of 1.4 and 2. Cold

treatment and cryogenic treatment of tools made of these steels can increase the pick lifelength by a factor of 1.1 to 2 compared with standard heat treatment.

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